

Operations Research, Central Exercise 4

Duality

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Recap: Standard Primal-Dual Pair

A linear program often has two equivalent viewpoints:

- the primal problem:

"How do I choose activities to maximize profit?"

- the dual problem:

"What is the minimum value of resources needed to support that profit?"

Primal (P)

$$\begin{array}{ll}\max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0\end{array}$$

Dual (D)

$$\begin{array}{ll}\min & b^T y \\ \text{s.t.} & A^T y \geq c \\ & y \geq 0\end{array}$$

- ▶ primal constraints $\leftrightarrow$ dual variables
- ▶ primal variables $\leftrightarrow$ dual constraints
- ▶ transpose coefficient matrix: $A \rightarrow A^T$

Recap - Primal $\rightarrow$ Dual Steps

1. Switch the **type of optimization**.
2. Introduce **as many dual variables as the number of primal constraints** (not counting the non-negativity constraints).
3. Define **as many dual constraints** (not counting the non-negativity constraints) **as the number of primal variables**.
4. Take the **transpose of the matrix of coefficients** (A^T).
5. **Switch the roles of the vector of coefficients** in the objective function and the vector of right-hand sides in the inequalities ($c^T \leftrightarrow b^T$).

SOB Method

Idea: Classify each variable and constraint as either *sensible* (S), *odd* (O) or *bizarre* (B).

Variable classification (independent of the type of optimization):

1. $x \geq 0 \leftarrow S$
2. $x \text{ free} \leftarrow O$
3. $x \leq 0 \leftarrow B$

Constraint classification:

Maximization

1. $Ax \leq b \leftarrow S$
2. $Ax = b \leftarrow O$
3. $Ax \geq b \leftarrow B$

Minimization

1. $Ax \geq b \leftarrow S$
2. $Ax = b \leftarrow O$
3. $Ax \leq b \leftarrow B$

SOB Method: Translation Rule

A **sensible**/**odd**/**bizarre** primal variable/constraint matches up with a **sensible**/**odd**/**bizarre** dual constraint/variable.

Primal variable type	Dual constraint type
<i>S</i>	sensible direction (<i>S</i>)
<i>O</i>	= (<i>O</i>)
<i>B</i>	bizarre direction (<i>B</i>)

Primal constraint type	Dual variable type
<i>S</i>	≥ 0 (<i>S</i>)
<i>O</i>	free (<i>O</i>)
<i>B</i>	≤ 0 (<i>B</i>)

For a **dual minimization** problem, the **sensible constraint direction is $\geq$** .

For a **dual maximization** problem, the **sensible constraint direction is $\leq$** .

Exercise D4.1 - Part 1

$$\begin{array}{llllll} \text{Max} & 4x_1 & - & 7x_2 & + & 8x_3 \\ \text{s.t.} & 3x_1 & & & + & 2x_3 \leq 18 \\ & 5x_1 & + & 7x_2 & - & x_3 \geq 5 \\ & -6x_1 & + & 4x_2 & + & 3x_3 = 30 \\ & x_1, x_2 \geq 0 \end{array}$$

Exercise D4.1 - Part 1 Solution

$$\begin{array}{ll}\text{Min} & 18y_1 + 5y_2 + 30y_3 \\ \text{s.t.} & 3y_1 + 5y_2 - 6y_3 \geq 4 \\ & 7y_2 + 4y_3 \geq -7 \\ & 2y_1 - y_2 + 3y_3 = 8 \\ & y_1 \geq 0 \\ & y_2 \leq 0 \\ & y_3 \text{ free}\end{array}$$

Exercise D4.1 - Part 2

$$\begin{array}{llllll} \text{Min} & x_1 & + & 2x_2 & - & 6x_3 \\ \text{s.t.} & 2x_1 & + & & + & 4x_3 = 12 \\ & & - & x_2 & + & 5x_3 \geq 10 \\ & -3x_1 & + & 3x_2 & + & x_3 \leq 20 \\ & x_1, x_2 & \geq & 0 & & \end{array}$$

Exercise D4.1 - Part 2 Solution

$$\begin{array}{ll}\text{Max} & 12y_1 + 10y_2 + 20y_3 \\ \text{s.t.} & 2y_1 - 3y_3 \leq 1 \\ & -y_2 + 3y_3 \leq 2 \\ & 4y_1 + 5y_2 + y_3 = -6 \\ & y_1 \text{ free} \\ & y_2 \geq 0 \\ & y_3 \leq 0\end{array}$$

Recap - Complementary Slackness

Primal (P)

$$\begin{array}{ll}\max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0\end{array}$$

Dual (D)

$$\begin{array}{ll}\min & b^T y \\ \text{s.t.} & A^T y \geq c \\ & y \geq 0\end{array}$$

Theorem (Complementary Slackness)

Let x^* and y^* be **feasible** solutions to the primal and the dual problem, respectively. The vectors x^* and y^* are **optimal** solutions for the two respective problems **if and only if**:

$$(A^T y^* - c)^T x^* = 0 \text{ and } (Ax^* - b)^T y^* = 0$$

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Interpretation:

- ▶ If $x_j^* > 0$, then the j -th dual constraint is binding.
- ▶ If $y_i^* > 0$, then the i -th primal constraint is binding.

Recap - Weak Duality

Primal (P)

$$\begin{array}{ll}\max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0\end{array}$$

Dual (D)

$$\begin{array}{ll}\min & b^T y \\ \text{s.t.} & A^T y \geq c \\ & y \geq 0\end{array}$$

Theorem (Weak Duality)

Let x' be a **feasible** solution of the primal problem and y' be a **feasible** solution of the dual problem. Then $c^T x' \leq b^T y'$.

Recap - Infeasibility / Unboundedness

Theorem (Weak Duality)

Let x' be a **feasible** solution of the primal problem and y' be a **feasible** solution of the dual problem. Then $c^T x' \leq b^T y'$.

Interpretation:

- ▶ Every dual feasible solution provides an **upper bound** on the primal objective value.
- ▶ Every primal feasible solution provides a **lower bound** on the dual objective value.

Consequences:

- ▶ If the primal is **unbounded above**, there cannot exist a dual feasible solution.
- ▶ If the dual is **unbounded below**, there cannot exist a primal feasible solution.

Recap - Infeasibility / Unboundedness

Theorem (Infeasibility / Unboundedness)

If the primal LP is unbounded, then the dual LP is infeasible.

If the dual LP is unbounded, then the primal problem is infeasible.

Self-Study Exercise

Mark (and argue!) the entries of the following table corresponding to possible outcomes in a primal/dual pair of linear programs:

		Dual		
		Finite optimum	Unbounded	Infeasible
Primal	Finite optimum			
	Unbounded			
	Infeasible			

Recap - Strong Duality

Theorem (Strong Duality)

If x^ is an **optimal** solution of the primal problem and y^* is an **optimal** solution of the dual problem, then $c^T x^* = b^T y^*$.*

Exercise D4.2 a)

$$\begin{array}{ll} \text{(P) Max} & 2x_1 + 3x_2 + x_3 \\ \text{s.t.} & 2x_1 + x_2 + x_3 \leq 4 \\ & 4x_1 + 2x_3 \leq 6 \\ & 5x_2 + 3x_3 \leq 8 \\ & x \in \mathbb{R}_{\geq 0}^3 \end{array} \qquad \begin{array}{ll} \text{(D) Min} & 4y_1 + 6y_2 + 8y_3 \\ \text{s.t.} & 2y_1 + 4y_2 \geq 2 \\ & y_1 + 5y_3 \geq 3 \\ & y_1 + 2y_2 + 3y_3 \geq 1 \\ & y \in \mathbb{R}_{\geq 0}^3 \end{array}$$

Specify all primal and dual complementary slackness conditions for (P) and (D).

Exercise D4.2 b)

Let $x^* = (3/2, 1, 0)^T$ be a primal point. Use the CS conditions to check whether x^* is an optimal solution for (P).

$$(2x_1 + x_2 + x_3 - 4) \cdot y_1 = 0$$

$$(4x_1 + 2x_3 - 6) \cdot y_2 = 0$$

$$(5x_2 + 3x_3 - 8) \cdot y_3 = 0$$

$$(2y_1 + 4y_2 - 2) \cdot x_1 = 0$$

$$(y_1 + 5y_3 - 3) \cdot x_2 = 0$$

$$(y_1 + 2y_2 + 3y_3 - 1) \cdot x_3 = 0$$

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$$(2y_1 + 4y_2 - 2) \cdot x_1 = 0$$

$$(y_1 + 5y_3 - 3) \cdot x_2 = 0$$

$$(y_1 + 2y_2 + 3y_3 - 1) \cdot x_3 = 0$$

Exercise D4.2 b) Solution

Given $x^* = \left(\frac{3}{2}, 1, 0\right)^T$.

Check primal feasibility:

$$4 \leq 4, \quad 6 \leq 6, \quad 5 \leq 8$$

Hence, x^* is feasible.

Since the third primal constraint is non-binding:

$$5x_2 + 3x_3 < 8$$

Complementary slackness implies: $y_3 = 0$.

Since $x_1 > 0$ and $x_2 > 0$: $2y_1 + 4y_2 = 2$ and $y_1 + 5y_3 = 3$.

Using $y_3 = 0$: $y_1 = 3, y_2 = -1$

But dual feasibility requires: $y_2 \geq 0$

Contradiction $\Rightarrow x^*$ is not optimal.

Exercise D4.2 c)

Is the dual problem (D) unbounded, infeasible or does it have at least one finite optimal solution? Answer this question without solving the dual problem. Justify your answer.

Exercise D4.2 c) Solution

The primal problem has a feasible solution:

$$x^* = \left(\frac{3}{2}, 1, 0\right)^T$$

Hence, by weak duality, the dual problem cannot be unbounded.

Furthermore,

$$y = (1, 1, 1)^T$$

is feasible for the dual problem.

Therefore, the dual problem is not infeasible.

It follows that the dual problem has at least one finite optimal solution.

Exercise D4.3

Consider the following transportation problem: You are responsible for distributing food supplies from collection points $i \in I$ to various food banks $j \in J$. Each collection point $i \in I$ has a supply capacity of food $s_i \geq 0$, and each food bank $j \in J$ has a demand $d_j \geq 0$ for food. Let $c_{ij} \geq 0$ denote the cost of transporting one unit of food from collection point $i \in I$ to food bank $j \in J$.

Exercise D4.3 a)

Formulate an LP (P) to determine a cost-minimizing transportation plan. Use variables $x_{ij} \geq 0$ for all $i \in I, j \in J$, where x_{ij} denotes the amount of food transported from collection point i to food bank j . Please ensure that (1) the demand at each food bank is fully met, and (2) no more food is transported from a collection point than is available.

Exercise D4.3 a) Solution

Decision variable:

$$x_{ij} \geq 0 \quad \forall i \in I, j \in J$$

amount transported from collection point i to food bank j .

$$\begin{aligned} \text{(P) Min} \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i \in I} x_{ij} \geq d_j \quad \forall j \in J \quad (\text{demand met}) \\ & \sum_{j \in J} x_{ij} \leq s_i \quad \forall i \in I \quad (\text{supply capacity}) \\ & x_{ij} \geq 0 \quad \forall i \in I, j \in J. \end{aligned}$$

Exercise D4.3 b)

Formulate the dual problem (D) corresponding to (P).

Exercise D4.3 b) Solution

Rewrite the supply constraints as:

$$-\sum_{j \in J} x_{ij} \geq -s_i \quad \forall i \in I.$$

Introduce dual variables:

$$p_j \geq 0 \quad \forall j \in J$$

for the demand constraints, and

$$\pi_i \geq 0 \quad \forall i \in I$$

for the rewritten supply constraints.

$$\begin{aligned} \text{(D) Max} \quad & \sum_{j \in J} d_j p_j - \sum_{i \in I} s_i \pi_i \\ \text{s.t.} \quad & p_j - \pi_i \leq c_{ij} \quad \forall i \in I, j \in J \\ & p_j, \pi_i \geq 0 \quad \forall j \in J, i \in I. \end{aligned}$$

Exercise D4.3 c)

The dual problem can be interpreted as the optimization problem of an external **transport company**. What are the meanings of the dual variables in this context?

Exercise D4.3 c) Interpretation

- ▶ π_i : purchase price at collection point i
- ▶ p_j : selling price at food bank j

The transport company wants to maximize:

$$\sum_{j \in J} d_j p_j - \sum_{i \in I} s_i \pi_i$$

subject to:

$$p_j - \pi_i \leq c_{ij}$$

Otherwise, transporting directly would be cheaper.

Notes

Notes